SDEV 2401

Rapid Backend App Development

Files and Media - 1

A LEADING POLYTECHNIC COMMITTED TO YOUR SUCCESS

Expectations - What I expect from you

- No Late Assignments

- No Cheating

- Be a good classmate

- Don’t waste your time

- Show up to class

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Agenda

On the right is what we will cover today.

- Static vs Media Files
- Static Files Setup
- Media Files Setup
- Model Example for Image Uploads
- Handling Image Uploads in Forms
- Displaying Uploaded Images
- Summary

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Static vs Media Files

There are two main types of files in a web application:

- **Static Files**: Files like CSS, JS, or icons used in your app's design.
- **Media Files**: Files uploaded by users, such as resumes or avatars.

Example:

- Static: `/static/icons/app-icon.svg`
- Media: `/media/uploads/user123/resume.pdf`
- Static files are bundled with your app and rarely change.
- Media files are dynamic and user-generated.
- Django needs different settings to serve each type.

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Static Files Setup

Let's talk about setting up static files in Django.

settings.py

```
STATIC_URL = '/static/'
STATICFILES_DIRS = [BASE_DIR / 'static_assets']
```

- Place your CSS/JS/images in `static_assets/` (you can name it anything).

Template Usage

```
{% load static %}
<link rel="stylesheet" href="{% static 'css/site.css' %}">
```

- `STATICFILES_DIRS` can point to any folder you want for static files.
- `{% static %}` helps Django find the right file path in templates.
- You'll need to run `python manage.py collectstatic` in to gather static files for production.

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Media Files Setup

settings.py

```
MEDIA_URL = '/media/'
MEDIA_ROOT = BASE_DIR / 'user_uploads'
```

- User uploads go in `user_uploads/`.

urls.py

```
from django.conf import settings
from django.conf.urls.static import static

# ... your project url patterns ...

if settings.DEBUG:
    urlpatterns += static(settings.MEDIA_URL, document_root=settings.MEDIA_ROOT)
```

- `MEDIA_ROOT` is where uploaded files are stored.
- The `static()` helper serves media files in development.

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Model Example for Image Uploads

models.py

```
from django.db import models
from django.conf import settings

class Photo(models.Model):
    user = models.ForeignKey(settings.AUTH_USER_MODEL, on_delete=models.CASCADE)
    image = models.ImageField(upload_to='photos/')
    uploaded_at = models.DateTimeField(auto_now_add=True)
```

- `ImageField` is used for image uploads and requires Pillow.
- `upload_to` sets the subfolder inside `MEDIA_ROOT`.
- `auto_now_add` records when the image was uploaded.

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Handling Image Uploads in Forms

forms.py

```
class PhotoForm(forms.ModelForm):
    class Meta:
        model = Photo
        fields = ['image']
```

views.py

```
def upload_photo(request):
    if request.method == 'POST':
        form = PhotoForm(request.POST, request.FILES)
        if form.is_valid():
            form.save()
            # Redirect or show success
    else:
        form = PhotoForm()
    return render(request, 'upload.html', {'form': form})
```

- Always use `request.FILES` for image uploads.
- The form will render a file input automatically.

Displaying Uploaded Images

Template Example

```
{% for photo in photos %}
    <img src="{{ photo.image.url }}" alt="User photo" width="200" />
{% endfor %}
```

- `.url` gives the path to the uploaded image.
- You can display images directly in your templates.
- Make sure your server is configured to serve media files in production.

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Summary

- Static files are for app assets; media files are for user uploads.
- Set up `STATIC_URL`, `STATICFILES_DIRS`, `MEDIA_URL`, and `MEDIA_ROOT` in settings.
- Use `ImageField` or `FileField` (we'll see `FileField` next class) in models for uploads.

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Example

Let’s go do an example together